Image Enhancement Assignment

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https://github.com/andrea13222/-CV-HW1-ImageEnhancement

1 Assignment requirements and objective

The assignment requires to develop a pipeline for enhancing degraded images by using basic image filtering techniques. The set of images (that could have been from 3 to 5 elements) should have some grade of condition that alters the details such as poor visibility, non uniform illumination or low contrast. The images has to be degraded and then enhanced to compare the results side by side and see the differences in details.

1.1 Images description and degradation

The actual set is composed of 3 images all taken by myself, each one of them contains some condition that the assignment required:

Figure 1: A metallic jar in front of a window

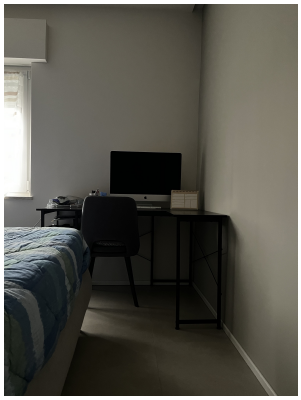

Figure 2: A bedroom with a workstation

Figure 3: The view of the road on a cloudy day

The images share low details respectively of the jar's decor in the first image, the underneath of the workstation in the second image and some details of the road and its surroundings in the third image To have a good comparison after

enhancing the images, they had to be degraded in order to get an enhancement that could potentially be the result of approaching the original photo, which serves as a benchmark that wouldn't normally exist in real-world scenarios. To degrade the images it has been lowered the contrast (to 0.45) and added gaussian noise ($\sigma = 12$).

2 Pipeline development

After generating the degraded images, four approaches have been applied and compared: the baseline, linear contrast stretching, histogram equalization and bilateral filtering. All methods use the same degraded images and fixed parameters to ensure a fair comparison.

2.1 Enhancement methods

The baseline consists of the degraded image without any further processing and is used as a reference. Linear contrast stretching increases the contrast by expanding the intensity range. The image is converted from BGR to CIE-Lab colour space and the operation is applied only to the L channel, keeping the colour channels unchanged. The 1st and 99th percentiles are used as minimum and maximum values to avoid extreme pixel values. Histogram equalization is also applied to the L channel in CIE-Lab, using OpenCV's `equalizeHist()` function. It redistributes the intensity values to improve contrast. However, it can also amplify noise and produce unnatural results. The fourth method is bilateral filtering, applied directly to the BGR image using a diameter of 7, sigmaColor of 50 and sigmaSpace of 50. It reduces noise while trying to preserve edges.

3 Evaluation

3.1 Metrics and quantitative results

The evaluation is performed by comparing each processed image with its original version, which acts as a reference for the synthetic degradation. In real-world situations, a clean reference image may not always be available. Two metrics are used: Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR). MSE measures the average squared pixel difference, while PSNR expresses the similarity in decibels. A lower MSE and a higher PSNR indicate a result closer to the original image. Below the table of the results:

Image	Method	PSNR_dB	MSE
img1	Baseline	13.37	2991.43
img1	Linear_Stretching	20.24	615.61
img1	Hist_Equalization	13.88	2661.02
img1	Bilateral_Filter	13.48	2918.21
img2	Baseline	15.85	1690.62
img2	Linear_Stretching	20.2	620.57
img2	Hist_Equalization	12.87	3354.66
img2	Bilateral_Filter	16.19	1562.61
img3	Baseline	15.29	1922.17
img3	Linear_Stretching	21.39	471.93
img3	Hist_Equalization	17.38	1187.57
img3	Bilateral_Filter	15.55	1811.28

The results show that linear contrast stretching provides the best numerical performance for all three images. It achieves the highest PSNR and the lowest MSE in every case. The best overall result is obtained on img3, with a PSNR of 21.39 dB and an MSE of 471.93.
A higher PSNR does not mean a better visual result. linear stretching achieves the highest PSNR across all images because its pixel values stay close to the original. The histogram equalization on img2 drops to 12.87 dB (worse than the 15.85 dB baseline) because it heavily shifts the overall brightness. However, the equalized image is visually much more informative, as it is the only one that makes the chair and the area under the desk visible.

4 Visual comparison and discussion

For each image, a side-by-side comparison has been created, showing the original image, its degraded version and the result obtained with each method (Baseline, Bilateral, Histogram eualization, Linear contrast stretching). The results show that in the first image the best method is the histogram equalization by adding details of the jar's decor. However, it amplifies gaussian noise, making flat areas look grainy. For the second one the linear stretching is the better one by adding details of underneath the desk. Histogram equalization brightens better the dark area under the desk, but makes the smooth wall noisy. And finally the last image both histogram and linear are good but the histogram shows better contrast on the clouds. The bilateral filter only removes noise while keeping edges sharp, but it does not fix the low contrast. Because of this, it looks almost identical to the degraded image, giving only a tiny improvement in PSNR. Each image includes its original state, the degraded one and the enhanced result:

Baseline, Bilateral, Histogram equalization and Linear stretching of img1

Baseline, Bilateral, Histogram equalization and Linear stretching of img2

Baseline, Bilateral, Histogram equalization and Linear stretching of img3